PROJECT CONCEPT NOTE

MindWell AI: Intelligent Mental Health Screening & Early Health Guidance for Youth

United Nations Sustainable Development Goal 3: Good Health and Well-being (Targets 3.4 & 3.d)

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1. Executive Summary

The contemporary youth demographic faces an unprecedented surge in psychological distress, academic anxiety, and depressive symptoms. Concurrently, prevailing societal stigma, high therapeutic costs, and severe counselor shortages hinder timely intervention. **MindWell AI** is an open-access, web-based digital health intervention engineered in direct alignment with UN SDG 3. By combining conversational large language models with validated psychometrics (PHQ-9 and GAD-7), somatic down-regulation biofeedback tools, and early symptom triage, MindWell AI provides accessible, confidential, and evidence-informed primary wellness support.

2. Problem Statement

> *"Lack of mental health support among youth results in stress and anxiety. This project aims to create an AI-powered chatbot for mental wellness check-ins and guidance."*

Key Systemic Challenges:

- **Stigma & Concealment:** Over 70% of struggling adolescents report avoiding institutional clinics due to fear of academic judgment or parental reprimand.
- **Provider Shortage:** The global psychiatrist-to-youth ratio remains below 1 per 100,000 in developing countries, leading to months-long waitlists.
- **Somatic Progression:** Chronic untreated stress manifests into physiological ailments (tension headaches, insomnia, digestive issues), impairing educational continuity.
- **Crisis Gap:** Traditional appointment systems fail during acute moments of panic or nocturnal despair when students need immediate grounding.

3. Proposed Solution & Core Capabilities

MindWell AI bridges this access gap through a multi-faceted digital wellness architecture:

- **Empathetic AI Companion:** Context-aware conversational agent for active listening, cognitive reframing, and exam de-stressing.
- **Standardized Clinical Screeners:** Validated PHQ-9 (Depression) & GAD-7 (Anxiety) scoring with instant risk tier classification.
- **AI Early Disease & Symptom Triage (SDG 3 Target 3.d):** Evaluates reported physical complaints against red-flag emergencies and recommends appropriate care levels.
- **Somatic Regulation Studio:** Real-time visual 4-7-8 and Box Breathing pacers, alongside a 5-4-3-2-1 sensory grounding module.
- **Instant Crisis Protocol:** Immediate heuristic safety detection connecting users to certified free 24/7 national hotlines (Tele-MANAS, KIRAN, 988, 116 123).

4. Technical Architecture & Tech Stack

Built with **Next.js 14 App Router**, **TypeScript**, and **Tailwind CSS** for instant serverless deployment on **Vercel**. The system employs a **Dual-Engine Architecture**: live Google Gemini 1.5 Flash / OpenAI inference when keys are provided, paired with an offline heuristic clinical triage engine to guarantee 100% uptime with zero 500 errors.

5. Ethical, Privacy & Clinical Guardrails

- **Explicit Disclaimers:** The application provides educational triage and self-care coaching, not formal psychiatric diagnosis.
- **Zero-PII Storage:** No personal tracking cookies or health record storage; session journals are kept strictly on the user's browser device.
- **Rapid Escalation:** Suicidal ideation flags bypass all normal bot chat and trigger emergency helpline cards immediately.

Submitted for: AI Project Evaluation (SDG 3: Good Health & Well-being) | **Repository:** github.com/GARVITBADRAHAN/ai-health-checker-